

KHOSEDA-KHARDSKY, Russian Federation

WMO: 232190

Lat: **67.033N**Lon: **59.400E**Elev: **272**StdP: **14.55**Time Zone: **3.00 (E03)**Period: **94-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.2	-39.9	-48.6	0.3	-44.0	-43.9	0.4	-39.0	33.6	5.7	30.2	9.5	1.6	0	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	79.0	65.7	74.9	63.4	70.2	60.8	67.4	77.4	64.7	72.9	61.8	68.7	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.7	72.8	61.0	81.0	70.0	58.4	73.7	66.6	32.0	77.8	30.0	73.7	27.9	68.5	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	21.5	18.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-47.2	71.1	5.3	1.9	-51.1	72.5	-54.2	73.6	-57.2	74.6	-61.1	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	-6.5	-3.1	6.2	20.2	33.6	49.0	57.0	48.9	40.5	27.4	9.9	2.0		
	DBStd	25.28	18.17	18.87	15.65	13.42	11.37	9.92	8.83	6.25	7.69	10.76	17.19	19.20		
	HDD50	9968	1752	1488	1358	894	522	137	33	97	298	699	1202	1488		
	HDD65	15031	2217	1908	1823	1344	975	482	277	502	734	1164	1652	1953		
	CDD50	446	0	0	0	0	13	108	249	62	14	0	0	0		
	CDD65	33	0	0	0	0	0	4	28	1	0	0	0	0		
	CDH74	454	0	0	0	0	7	54	374	19	0	0	0	0		
	CDH80	90	0	0	0	0	0	5	82	3	0	0	0	0		
(14) Wind	WSAvg	8.2	7.9	8.1	8.4	8.7	8.9	8.7	7.6	7.8	7.4	8.3	8.0	9.1		
(15) Precipitation	PrecAvg	17.9	1.2	0.9	0.8	0.8	1.3	1.6	2.1	2.6	2.0	1.9	1.3	1.2		
	PrecMax	25.8	2.3	1.8	1.5	2.0	2.1	3.9	6.3	6.0	4.6	3.5	2.2	2.5		
	PrecMin	11.4	0.5	0.0	0.2	0.1	0.0	0.4	0.1	0.6	0.0	0.9	0.6	0.4		
	PrecStd	3.5	0.5	0.4	0.3	0.4	0.5	0.8	1.2	1.1	0.9	0.7	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	32.7	39.2	51.3	74.5	78.9	86.2	76.3	66.9	49.5	36.2	33.3		
		MCWB	28.5	31.6	35.6	44.5	60.9	64.8	68.4	64.9	58.3	47.7	35.9	32.6		
	2%	DB	26.5	29.6	33.8	43.2	64.3	75.3	81.8	70.0	60.5	45.2	33.2	31.6		
		MCWB	25.6	28.6	32.0	38.5	53.3	63.3	67.5	59.0	54.1	44.0	32.6	31.0		
	5%	DB	23.7	26.7	31.8	39.7	56.4	70.8	78.1	65.1	56.1	42.0	32.1	28.8		
		MCWB	22.8	25.7	30.7	36.3	49.0	61.4	65.6	57.7	51.7	41.2	31.6	27.9		
	10%	DB	17.6	22.9	28.4	36.8	49.8	66.6	73.8	61.2	52.6	39.9	30.2	25.7		
		MCWB	16.7	21.9	26.9	34.5	44.8	58.3	63.8	56.0	49.4	39.2	29.5	24.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.2	31.8	36.2	44.6	61.1	68.6	70.9	66.4	60.5	48.1	35.8	33.0		
		MCDB	30.2	32.6	38.8	51.0	72.9	76.3	80.4	72.7	65.4	49.3	36.5	33.1		
	2%	WB	25.6	28.6	32.4	39.1	54.5	65.0	68.7	61.0	55.7	44.2	32.9	31.0		
		MCDB	26.4	29.3	33.4	42.1	63.6	72.2	79.8	66.7	59.0	45.0	33.1	31.4		
	5%	WB	22.7	25.7	30.7	36.8	49.5	62.1	66.7	58.5	52.1	41.3	31.6	27.9		
		MCDB	23.8	26.6	31.8	39.2	56.1	69.6	76.8	64.0	55.3	41.9	32.0	28.6		
	10%	WB	16.8	21.9	27.1	34.8	44.9	59.1	64.3	55.9	49.6	39.2	29.3	24.8		
		MCDB	17.6	22.8	28.2	36.6	49.8	66.5	72.4	60.8	52.4	39.9	30.0	25.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	15.1	20.0	16.9	14.5	16.7	18.7	14.6	12.7	9.3	12.4	13.8		
		MCDBR	13.3	10.7	13.6	16.1	26.1	23.7	26.2	24.1	18.3	9.1	8.0	10.8		
		MCWBR	13.1	10.5	12.7	13.1	17.9	15.0	13.9	15.1	12.7	8.5	8.4	11.0		
	5% WB	MCDBR	13.6	11.1	13.2	14.2	24.8	22.7	23.8	21.5	15.6	8.7	8.7	11.6		
		MCWBR	13.5	11.0	12.5	12.0	17.6	15.8	14.1	14.9	11.7	8.4	8.8	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.215	0.265	0.304	0.332	0.345	0.355	0.386	0.365	0.323	0.277	0.207	N/A		
	taud		1.885	2.014	2.060	2.088	2.228	2.432	2.355	2.395	2.467	2.228	1.925	N/A		
	Ebn at Noon		42	170	229	257	268	270	255	248	233	171	41	0		
	Edh at Noon		4	18	29	37	36	31	32	28	20	14	4	0		
(44) All-Sky Solar Radiation	RadAvg		16	162	637	1223	1612	1587	1556	974	545	174	35	0		
	RadStd		2	28	89	111	156	164	243	111	63	28	5	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+2.30	N/A	N/A	N/A	N/A	N/A	N/A	-731	-806	N/A	N/A

Nomenclature: See separate page